BANK CHURN ANALYSIS

use bank;

select * from bank_data;

CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
15634602	Hargrave	619	France	Female	42	2	0	1	1	1	101348.88	1
15647311	Hill	608	Spain	Female	41	1	83807.86	1	0	1	112542.58	0
15619304	Onio	502	France	Female	42	8	159660.8	3	1	0	113931.57	1
15701354	Boni	699	France	Female	39	1	0	2	0	0	93826.63	0
15737888	Mitchell	850	Spain	Female	43	2	125510.82	1	1	1	79084.1	0
15574012	Chu	645	Spain	Male	44	8	113755.78	2	1	0	149756.71	1
15592531	Bartlett	822	France	Male	50	7	0	2	1	1	10062.8	0
15656148	Obinna	376	Germany	Female	29	4	115046.74	4	1	0	119346.88	1
15792365	He	501	France	Male	44	4	142051.07	2	0	1	74940.5	0
15592389	H?	684	France	Male	27	2	134603.88	1	1	1	71725.73	0
15767821	Bearce	528	France	Male	31	6	102016.72	2	0	0	80181.12	0
15737173	Andrews	497	Spain	Male	24	3	0	2	1	0	76390.01	0

-- How many total customers are present in the dataset?

select count(*) as total_customers from bank_data;

total_customers
10000

-- How many customers have exited the bank?

select count(*) as total_customer_exited from bank_data
where exited = 1;

total_customer_exited
2037

-- How many customers are still active (not exited)?

select count(*) as still_active from bank_data
where exited <> 1;

still_active
7963

-- What is the average credit score of customers?

select avg(creditscore) as avg_credit_score from bank_data;

avg_credit_score
650.5288

-- What is the minimum and maximum age of customers?

select min(age) as minimum_age, max(age) as maximum_age from bank_data;

minimum_age	maximum_age
18	92

-- How many customers belong to each geography?
select geography, count(*) from bank_data
group by geography;

geography	count(*)
France	5014
Spain	2477
Germany	2509

-- How many male and female customers are there?
select gender, count(*) as gender_count from bank_data
group by gender ;

gender	gender_count
Female	4543
Male	5457

-- What is the average balance of all customers?
select avg(balance) from bank_data;

avg(balance)
76485.88928799961

select avg(balance) from bank_data
where exited = 1;

avg(balance)
91108.53933726063

select gender, avg(balance) from bank_data
group by gender;

gender	avg(balance)
Female	75659.36913933513
Male	77173.97450613906

select gender, avg(balance) from bank_data
where exited = 1
group by gender;

gender	avg(balance)
Female	89036.63935908697
Male	93736.48374164799

-- How many customers have a credit card?
select count(*) as total from bank_data;

total
10000

```
select count(*)  as Exited from bank_data
where exited = 1;
```

Exited
2037

```
select gender, count(*) from bank_data
group by gender;
```

gender	count(*)
Female	4543
Male	5457

-- How many customers are active members of the bank?

```
select count(*) from bank_data
where isactivemember = 1;
```

active_member
5151

-- What is the average estimated salary of customers?

```
select round((EstimatedSalary),2) as Avg_Salary from bank_data;
```

Avg_Salary
101348.88
112542.58
113931.57
93826.63
79084.1
149756.71
10062.8
119346.88
74940.5
71725.73
80181.12
76390.01

-- How many customers have more than one product?

```
select count(*) as more_than_one_product from bank_data
where NumOfProducts > 1;
```

more_than_one_product
4916

-- What is the distribution of customers by tenure?

```
select Tenure,count(*) from bank_data
group by Tenure
order by Tenure;
```

Tenure	count(*)
0	413
1	1035
2	1048
3	1009
4	989
5	1012
6	967
7	1028
8	1025
9	984
10	490

-- How many customers have zero balance?

```
select count(*) as zero_balance from bank_data
where balance = 0;
```

zero_balance
3617

-- What is the average age of customers who exited?

```
select avg(age) as avg_age from bank_data;
```

avg_age
38.9218

-- What is the average credit score of customers who exited vs those who stayed?

```
select case
when Exited = 1 then 'Exited '
else 'Stayed'
end as Exited, avg(CreditScore) as Avg_Credit_Score
from bank_data
group by exited;
```

Exited	Avg_Credit_Score
Exited	645.3515
Stayed	651.8532

-- Which geography has the highest number of customers?

```
select Geography, count(*) as Highest_no_customer from bank_data
group by Geography
order by count(*) desc;
```

Geography	Highest_no_customer
France	5014
Germany	2509
Spain	2477

-- Which geography has the highest churn rate?

```
select Geography, count(*) as total_customer,
avg(Exited) * 100 as churn_rate
from bank_data
group by Geography;
```

Geography	total_customer	churn_rate
France	5014	16.1548
Spain	2477	16.6734
Germany	2509	32.4432

-- What is the average balance for customers in each geography?

```
select Geography, avg(Balance) as Avg_balance from bank_data
group by Geography;
```

Geography	Avg_balance
France	62092.6365157559
Spain	61818.14776342349
Germany	119730.11613391782

-- How does churn vary by gender?

```
select gender, round(avg(Exited) * 100 ,2) as churn from bank_data
group by gender;
```

gender	churn
Female	25.07
Male	16.46

-- What is the average age of customers who exited vs those who stayed?

```
select case
when Exited = 1 then 'Exited '
else 'Stayed'
end as Exited, avg(age) as avg_age from bank_data
group by exited;
```

Exited	avg_age
Exited	44.8380
Stayed	37.4084

-- What is the average age of customers who exited vs those who stayed group by gender?

```
select case
when Exited = 1 then 'Exited '
else 'Stayed'
end as Exited, gender, avg(age) as avg_age from bank_data
group by exited, gender;
```

Exited	gender	avg_age
Exited	Female	44.7849
Stayed	Female	37.3825
Exited	Male	44.9053
Stayed	Male	37.4277

-- What is the average estimated salary by geography?

```
select geography,gender, round(avg(EstimatedSalary),2) as avg_salary from bank_data
group by geography,gender
order by geography;
```

geography	gender	avg_salary
France	Female	99564.25
France	Male	100174.25
Germany	Female	102446.42
Germany	Male	99905.03
Spain	Female	100734.11
Spain	Male	98425.69

-- What is the churn rate for customers who have a credit card vs those who don't?

```
select case
when HasCrCard = 1 then 'Has a Card'
else 'No Card'
end as
HasCrCard,Gender, count(*),
count(*) * 100 / (select count(*) from bank_data) as Percentage from bank_data
group by HasCrCard, gender;
```

HasCrCard	Gender	count(*)	Percentage
Has a Card	Female	3192	31.9200
No Card	Female	1351	13.5100
Has a Card	Male	3863	38.6300
No Card	Male	1594	15.9400

-- What is the churn rate for active vs inactive members?

```
select case
when IsActiveMember = 1 then 'Active Member'
else 'Non Active'
end as churn_rate,
count(*) * 100 / (select count(*) from bank_data) as churn_rate_in_Percent,
count(*) from bank_data
group by IsActiveMember;
```

churn_rate	churn_rate_in_Percent	count(*)
Active Member	51.5100	5151
Non Active	48.4900	4849

-- What is the average number of products used by exited customers?

```
select case
when Exited = 1 then 'Exited customer'
else 'not exited'
end as Exited_customer, avg(NumOfProducts) as avg_num_of_product from bank_data
where Exited = 1;
```

Exited_customer	avg_num_of_product
Exited customer	1.4752

-- Which tenure group has the highest churn rate?

```
select Tenure, count(*) as churn,
count(*) * 100 / (select count(*) from bank_data where exited = 1) as churn_rate from bank_data
where exited = 1
group by tenure
order by tenure desc;
```

Result Grid | Filter Rows:

Tenure	churn	churn_rate
10	101	4.9583
9	213	10.4566
8	197	9.6711
7	177	8.6892
6	196	9.6220
5	209	10.2602
4	203	9.9656
3	213	10.4566
2	201	9.8675
1	232	11.3893
0	95	4.6637

-- What is the average balance for male vs female customers?

```
select gender,avg(balance) as Avg_balance from bank_data
group by gender;
```

Result Grid | Filter Rows:

gender	Avg_balance
Female	75659.36913933513
Male	77173.97450613906

-- Which age group has the highest number of exits?

```
select case
when age >= 18 and age < 25 then 'Young Age'
when age >= 25 and age < 40 then 'Adult Age'
else 'Old Age'
end as Age_Group  ,count(*) as Exited from bank_data
where exited = 1
group by Age_Group;
```

Result Grid | Filter Rows

Age_Group	Exited
Old Age	1440
Adult Age	557
Young Age	40

-- What is the average credit score by geography?

```
select geography, avg(CreditScore) as Avg_credit_score from bank_data
group by geography;
```

Result Grid | Filter Rows:

geography	Avg_credit_score
France	649.6683
Spain	651.3339
Germany	651.4536

-- what is the average credit score by geography and by Age_group

with age_groups as (select case
when age >= 18 and age < 25 then 'Young Age'
when age >= 25 and age < 40 then 'Adult Age'
else 'Old Age'
end as Age_Group, CustomerID from bank_data)

Age_Group	geography	Avg_credit_score
Adult Age	France	649.2456
Adult Age	Germany	656.4627
Adult Age	Spain	650.8491
Old Age	France	649.1707
Old Age	Spain	651.9549
Old Age	Germany	646.4167
Young Age	Spain	651.8387
Young Age	France	658.7890
Young Age	Germany	644.8438

select a.Age_Group, b.geography, avg(b.CreditScore) as Avg_credit_score from bank_data b join
age_groups a on b.CustomerID = a.customerID
group by b.geography, a.Age_Group
order by a.age_group;

Age_Group	geography	Avg_credit_score
Adult Age	France	649.2456
Adult Age	Germany	656.4627
Adult Age	Spain	650.8491
Old Age	France	649.1707
Old Age	Spain	651.9549
Old Age	Germany	646.4167
Young Age	Spain	651.8387
Young Age	France	658.7890
Young Age	Germany	644.8438

-- What percentage of customers have more than 2 products?

select NumOfProducts, count(*) * 100 / (select count(*) from bank_data) as percentage from Bank_data
where NumOfProducts > 2
group by NumOfProducts;

NumOfProducts	percentage
3	2.6600
4	0.6000

-- Find the top 10 customers with the highest balance.

select surname,
max(balance) over (order by balance desc) as max_balance from bank_data;

surname	max_balance
Lo	250898.09
To Rot	250898.09
Haddon	250898.09
McIntosh	250898.09
Shaw	250898.09
Dilke	250898.09
Chia	250898.09
Moore	250898.09
Thomson	250898.09
Mao	250898.09
Udinesi	250898.09
Macartney	250898.09

-- using window rank function

```
select surname, balance,
rank() over (order by balance desc) as ranking
from bank_data
limit 10;
```

surname	balance	Ranking
Lo	250898.09	1
To Rot	238387.56	2
Haddon	222267.63	3
McIntosh	221532.8	4
Shaw	216109.88	5
Dilke	214346.96	6
Chia	213146.2	7
Moore	212778.2	8
Thomson	212696.32	9
Mao	212692.97	10

-- Find the average salary of customers grouped by geography and gender.

```
select geography, gender,round( avg(EstimatedSalary) ,2) from bank_data
group by geography, gender
order by geography;
```

Result Grid Filter Rows:

geography	gender	average
France	Female	99564.25
France	Male	100174.25
Germany	Female	102446.42
Germany	Male	99905.03
Spain	Female	100734.11
Spain	Male	98425.69

-- Identify customers whose credit score is below the average credit score.

```
select surname,
case
when CreditScore < 500 then 'Wrost'
when CreditScore between 500 and 700 then 'Good'
else 'Excellent'
End as Credit_Score from bank_data
where CreditScore < (select avg(CreditScore) from bank_data);
```

Result Grid | Filter Rows:

surname	Credit_Score
Hargrave	Good
Hill	Good
Onio	Good
Chu	Good
Obinna	Wrost
He	Good
Bearce	Good
Andrews	Wrost
Kay	Wrost
Chin	Good
Scott	Good
Goforth	Good

-- Rank customers based on their balance within each geography.

select * from (select surname,geography,
rank() over (partition by geography order by balance desc) as rank_by_salary
from bank_data) t
where rank_by_salary <=5;

surname	geography	rank_by_salary
To Rot	France	1
Moore	France	2
Mao	France	3
Udinesi	France	4
Macartney	France	5
Dilke	Germany	1
Thomson	Germany	2
Olisaemeka	Germany	3
Conti	Germany	4
Connor	Germany	5
Lo	Spain	1
Haddon	Spain	2

-- Find the top 5 customers with the highest estimated salary in each geography.

select * from (select surname, geography,
rank() over (partition by geography order by EstimatedSalary desc) as High_salary
from bank_data) t
where High_salary <=5;

surname	geography	High_salary
Moss	France	1
Adams	France	2
DeRose	France	3
Oluchukwu	France	4
Chibueze	France	5
Dyer	Germany	1
Hopkins	Germany	2
Mai	Germany	3
Lu	Germany	4
Chidozie	Germany	5
Lucciano	Spain	1
Gannon	Spain	2

-- Calculate the churn rate for each age group.

with age_groups as (select CustomerId, case
when age >= 18 and age <= 25 then 'Young'

```
when age >= 26 and age <= 40 then 'Adult'
else 'Old'
end as Age_Group from bank_data)
select a.Age_Group,
count(*) as total_customer,
sum(b.Exited) as chruned_customer,
avg(b.Exited) * 100 as chruned_rate
from bank_data b
join age_groups a
on b.customerid = a.customerid
group by a.Age_Group;
```

Age_Group	total_customer	chruned_customer	chruned_rate
Old	3581	1351	37.7269
Adult	5808	640	11.0193
Young	611	46	7.5286

-- Find customers whose balance is greater than the average balance of their geography.

```
select customerid, surname, geography,balance from bank_data b
where balance > (select avg(balance) from bank_data where geography = b.geography);
```

customerid	surname	geography	balance
15647311	Hill	Spain	83807.86
15619304	Onio	France	159660.8
15737888	Mitchell	Spain	125510.82
15574012	Chu	Spain	113755.78
15792365	He	France	142051.07
15592389	H?	France	134603.88
15767821	Bearce	France	102016.72
15643966	Goforth	Germany	143129.41
15737452	Romeo	Germany	132602.88
15736816	Young	Germany	136815.64
15728693	McWilliams	Germany	141349.43
15706552	Odinak...	France	85311.7

```
select customerid, surname, geography,balance from (
select *,
row_number() over (partition by geography order by balance desc) as rank_balance,
avg(balance) over (partition by geography ) as average_balance from bank_data
) t
where balance > average_balance and rank_balance <= 10;
```

customerid	surname	geography	balance
15715622	To Rot	France	238387.56
15769818	Moore	France	212778.2
15780212	Mao	France	212692.97
15690589	Udinesi	France	212314.03
15671256	Macartney	France	211774.31
15736420	Macdonald	France	210433.08
15709920	Burke	France	208165.53
15627971	Coates	France	206663.75
15784180	Ku	France	206329.65
15664498	Golovanov	France	205962
15599131	Dilke	Germany	214346.96
15620268	Thomson	Germany	212696.32

-- Find customers with the highest credit score in each geography.

```
select customerid,geography, CreditScore, surname from (
select *,
```

```
row_number() over (partition by geography order by CreditScore desc) as Hig_CR,
max(CreditScore) over (partition by geography) as maximum_CR
from bank_data
) ta
where Hig_CR <= 10;
```

customerid	geography	CreditScore	surname
15682868	France	850	Elliott
15709256	France	850	Glover
15708904	France	850	Yermakova
15621457	France	850	Chu
15669491	France	850	Cruz
15739847	Germany	850	Sadlier
15629078	Germany	850	Matthias
15595221	Germany	850	Trevisano
15607178	Germany	850	Welch
15602010	Germany	850	Zikoranau...
15719958	Germany	850	Degtyarev
15686164	Germany	850	Maclean

-- Identify customers who have more than 2 products but still exited.

```
select customerid,surname, NumOfProducts from bank_data
where NumOfProducts >1 and Exited = 1;
```

customerid	surname	NumOfProducts
15619304	Onio	3
15574012	Chu	2
15656148	Obinna	4
15589475	Azikiwe	3
15703793	Konovalova	4
15622897	Sharpe	3
15757535	Heap	3
15613854	Mauldon	2
15609618	Fanucci	2
15640905	Vasin	2
15574692	Pinto	2
15726931	Onwumelu	2

```
select NumOfProducts,
count(*) as total_customer,
sum(Exited) as total_exited,
avg(Exited) * 100 as churn_rate
from bank_data
group by NumOfProducts;
```

NumOfProducts	total_customer	total_exited	churn_rate
1	5084	1409	27.7144
3	266	220	82.7068
2	4590	348	7.5817
4	60	60	100.0000

-- Find the average balance of the top 20% customers with the highest salaries.

```
select avg(balance) as average_balance
from (
select balance,
ntile(5) over (order by EstimatedSalary desc) as salary_group
from bank_data
) t
where salary_group = 1;
```

	average_balance
▶	77052.09980999997

-- Find the percentage of customers who exited in each geography.

SELECT geography,
COUNT(*) AS exited_customers,
COUNT(*) * 100.0 / (
  SELECT COUNT(*)
  FROM bank_data b2
  WHERE b2.geography = b1.geography
) AS exited_percent
FROM bank_data b1
WHERE exited = 1
GROUP BY geography;

	geography	exited_customers	exited_percent
▶	France	810	16.15477
	Spain	413	16.67340
	Germany	814	32.44320

-- Rank customers by credit score within each gender.

select gender,customerid, surname,CreditScore, RowNumber, Rnk,DenseRnk from (select *,
row_number() over (partition by gender order by CreditScore desc) as RowNumber,
rank() over (partition by gender order by CreditScore desc) as Rnk,
dense_rank() over (partition by gender order by CreditScore desc) as DenseRnk
from bank_data) t
where RowNumber <= 10;

	gender	customerid	surname	CreditScore	RowNumber	Rnk	DenseRnk
▶	Female	15798615	Wan	850	1	1	1
	Female	15647800	Greco	850	2	1	1
	Female	15672574	Uspenskaya	850	3	1	1
	Female	15682868	Elliott	850	4	1	1
	Female	15709256	Glover	850	5	1	1
	Female	15708904	Yermakova	850	6	1	1
	Female	15800554	Perry	850	7	1	1
	Female	15604536	Vachon	850	8	1	1
	Female	15669491	Cruz	850	9	1	1
	Female	15572390	Huang	850	10	1	1
	Male	15815095	Burfitt	850	1	1	1
	Male	15745250	Simpson	850	2	1	1

-- Identify customers whose salary is above the overall average salary but still exited.

select Distinct customerid, surname, EstimatedSalary, rnk from (
select *,
row_number() over (partition by exited order by EstimatedSalary desc) as rnk,
avg(EstimatedSalary) over (partition by Exited) as avg_salary from bank_data
) t
where EstimatedSalary > avg_salary and Exited = 1 ;

Result Grid | Filter Rows:

customerid	surname	EstimatedSalary	rnk
15815656	Hopkins	199808.1	1
15661670	Chidozie	[illegible]725.39	2
15672152	Grant	[illegible]693.84	3
15661903	Hsia	199378.58	4
15755262	McDonald	199304.74	5
15624641	Kharlamova	199290.68	6
15707362	Yin	199273.98	7
15671591	Castiglione	198874.52	8
15699095	Chandler	198826.03	9
15585100	Rioux	198814.24	10
15754577	Boni	198810.65	11
15662751	Piazza	198798.44	12

-- Find the top 3 age groups with the highest churn rate.

```
Select age_group,
count(*) as total_customer,
sum(Exited) as total_exited,
avg(Exited) * 100 as churn_rate from (
select *,case
            when age between 18 and 25 then 'Young'
   when age between 26 and 40 then 'Adult'
   when age between 41 and 60 then 'Middle Age'
   else 'Senior'
end as Age_group from bank_data
) t
group by Age_group
order by churn_rate desc
limit 3;
```

Age_group	total_customer	total_exited	churn_rate
Middle Age	3117	1236	39.6535
Senior	464	115	24.7845
Adult	5808	640	11.0193

-- Calculate the cumulative balance of customers ordered by credit score.

```
select customerID, surname, geography,gender,age, creditScore, balance,
sum(balance) over (partition by geography order by CreditScore ) as Cumullative_balance
from bank_data;
```

customerID	surname	geography	gender	age	creditScore	balance	Cumullative_balance
15668309	Maslow	France	Female	40	350	111098.85	111098.85
15765173	Lin	France	Female	60	350	0	111098.85
15803202	Onyekachi	France	Male	51	350	0	111098.85
15612494	Panicucci	France	Female	44	359	128747.69	239846.54
15734711	Loggia	France	Male	42	373	0	239846.54
15804586	Lin	France	Female	46	376	0	239846.54
15615296	Rice	France	Male	39	405	0	239846.54
15583849	Ts'ai	France	Male	40	408	0	327719.93
15653251	Hickey	France	Female	84	408	87873.39	327719.93
15628155	Dike	France	Female	35	410	117183.74	570693.36
15603096	Lori	France	Male	33	410	125789.69	570693.36
15660475	Ndubueze	France	Female	54	411	0	630390.53

-- Find the top 5 customers with the highest balance in each geography.

```
select CustomerId, Surname, Geography, Balance from (
select *,
row_number () over (partition by geography order by balance desc) as rnk
from bank_data
) t
where rnk <= 5;
```

CustomerId	Surname	Geography	Balance
15715622	To Rot	France	238387.56
15769818	Moore	France	212778.2
15780212	Mao	France	212692.97
15690589	Udinesi	France	212314.03
15671256	Macartney	France	211774.31
15599131	Dilke	Germany	214346.96
15620268	Thomson	Germany	212696.32
15795298	Olisaemeka	Germany	206868.78
15745433	Conti	Germany	205770.78
15663888	Connor	Germany	204017.4
15757408	Lo	Spain	250898.09
15714241	Haddon	Spain	222267.63

-- Find customers whose balance is greater than the average balance of their geography.

```
select CustomerId, Surname, Geography, Balance, avg_balance from (
select *,
row_number() over (partition by Geography order by Geography) as rnk,
avg(balance) over (partition by geography order by Geography) as avg_balance
from bank_data
) t
where balance > avg_balance and rnk <= 10;
```

CustomerId	Surname	Geography	Balance	avg_balance
15611365	Fanucci	France	127661.69	62092.636515755934
15701972	Parsons	France	137179.39	62092.636515755934
15780362	Ferrari	France	119960.29	62092.636515755934
15753229	Genovese	France	127414.55	62092.636515755934
15669282	Uchechukwu	France	124266.86	62092.636515755934
15792726	Sung	France	127974.06	62092.636515755934
15569678	Cocci	Germany	166824.59	119730.11613391797
15775663	Otitodilichukwu	Germany	134729.99	119730.11613391797
15805260	Wood	Germany	143249.67	119730.11613391797
15747174	Hao	Germany	190298.89	119730.11613391797
15577831	Byrne	Germany	152532.3	119730.11613391797
15580935	Okechukwu	Germany	135962.4	119730.11613391797

-- Rank customers within each geography based on their credit score.

```
select CustomerId, Surname, Geography, Balance CreditScore, ro from (
select *,
row_number() over (partition by Geography order by CreditScore desc) as ro,
rank() over (partition by Geography order by CreditScore desc) as rnk,
dense_rank() over (partition by Geography order by CreditScore desc) as dens
from bank_data
) t
where ro <= 10;
```

CustomerId	Surname	Geography	CreditScore	ro
15743760	Davidson	France	131996.66	1
15647800	Greco	France	101266.51	2
15730579	Ward	France	169445.4	3
15634591	Saunders	France	73059.38	4
15798615	Wan	France	137301.87	5
15682868	Elliott	France	99816.46	6
15709256	Glover	France	0	7
15708904	Yermakova	France	0	8
15621457	Chu	France	96654.72	9
15669491	Cruz	France	157866.77	10
15739847	Sadlier	Germany	146756.68	1
15629078	Matthias	Germany	127258.79	2

-- Calculate the running total of balance ordered by credit score.

```
select customerID, surname, geography,gender,age, creditScore, balance,
sum(balance) over (partition by geography order by CreditScore ) as Cumullative_balance
from bank_data;
```

customerID	surname	geography	gender	age	creditScore	balance	Cumullative_balance
15668309	Maslow	France	Female	40	350	111098.85	111098.85
15765173	Lin	France	Female	60	350	0	111098.85
15803202	Onyekachi	France	Male	51	350	0	111098.85
15612494	Panicucci	France	Female	44	359	128747.69	239846.54
15734711	Loggia	France	Male	42	373	0	239846.54
15804586	Lin	France	Female	46	376	0	239846.54
15615296	Rice	France	Male	39	405	0	239846.54
15583849	Ts'ai	France	Male	40	408	0	327719.93
15653251	Hickey	France	Female	84	408	87873.39	327719.93
15628155	Dike	France	Female	35	410	117183.74	570693.36
15603096	Lori	France	Male	33	410	125789.69	570693.36
15660475	Ndubueze	France	Female	54	411	0	630390.53

```
select
   Geography,
   Gender,
   case
      when Age between 18 and 25 then 'Young'
      when Age between 26 and 40 then 'Middle Age'
      when Age between 41 and 60 then 'Adult'
      else 'Senior'
   end as Age_Group,

   COUNT(*) as Total_Customers,
   SUM(Exited) as Churned_Customers,

   ROUND(Avg(Exited) * 100.0 ,2) as Churn_Rate
from bank_data
group by Geography, Gender, Age_Group
order by Churn_Rate desc;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Geography	Gender	Age_Group	Total_Customers	Churned_Customers	Churn_Rate
Germany	Female	Adult	459	268	58.39
Germany	Female	Senior	53	27	50.94
Germany	Male	Adult	462	232	50.22
France	Female	Adult	670	274	40.90
Spain	Female	Adult	362	141	38.95
France	Male	Adult	764	219	28.66
Germany	Male	Senior	60	17	28.33
Spain	Male	Adult	400	102	25.50
France	Female	Senior	111	28	25.23
Germany	Female	Middle Age	613	142	23.16
Spain	Male	Senior	71	16	22.54
Spain	Female	Senior	49	9	18.37

SUMMERY :

The document performs **SQL-based analysis on a bank customer dataset (**bank_data) to understand **customer behavior and churn (customers leaving the bank)**. It uses **aggregations, CASE statements, CTEs, and window functions** to extract business insights.

DATASET OVERVIEW

- CustomerId
- Surname
- CreditScore
- Geography
- Gender
- Age
- Tenure
- Balance
- NumOfProducts
- HasCrCard (Yes = 1, No = 0)
- IsActiveMember (Yes = 1, No = 0)
- EstimatedSalary
- Exited (Yes = 1, No = 0)

Customer Demographics & Status: It determines the total number of customers , counts how many have exited vs how many remain active , and breaks down the customer base by geography and gender. It also identifies the minimum and maximum ages of the customers.

Financial Profiling: The queries calculate average credit scores, average account balances, and average estimated salaries. These financial metrics are further analyzed by grouping them by gender, geography, and whether the customer has exited or stayed.

Product Engagement: The analysis looks at customer engagement by counting how many customers possess a credit card, how many are considered active members, and how many utilize more than one bank product.

Churn Drivers: A significant portion of the document focuses on identifying churn rates across different segments. It examines how churn varies by geography , gender , credit card ownership , active membership status , and tenure groups. It also investigates which age groups have the highest number of exits.

Advanced Ranking and Segmentation: Using window functions, the analysis ranks top customers based on their balances and credit scores within specific geographic regions and genders. It also identifies specific customer profiles, such as customers with above average salaries who still chose to exit the bank.